DO NOW — RECALL

Lesson 087 · Algorithm Design and Testing Challenge

Name: _____ Class: _____ Date: _____

1. What is an invariant?

2. Why is sorting unnecessary for just finding one maximum?

3. What is nullable int?

4. Binary search for 8 in [2,4,6,8,10]. Give midpoint indexes and result.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A condition maintained while processing data.

2. Model answer

A single scan can find it in O(n).

3. Model answer

An int? that can be an integer or null.

4. Model answer

mid2=6, then mid3=8; found index3.

By the end, you can…

• Specify output when inputs are empty, and preserve correctness on all-negative arrays.

• Maintain invariant: best is the largest value among those inspected so far.

• Compare against alternative sorting-based solution: an O(n) scan avoids O(n log n) sort when only the maximum is needed.

Vocabulary — know these terms

maximum — Largest value in set.

invariant — Maintained fact during loop.

nullable — Type allowing absence/null.

complexity — Resource growth as input increases.

NEW LEARNING

Read, annotate and explain

Observe

Specify output when inputs are empty, and preserve correctness on all-negative arrays.

Explain

Maintain invariant: best is the largest value among those inspected so far.

Evaluate

Compare against alternative sorting-based solution: an O(n) scan avoids O(n log n) sort when only the maximum is needed.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

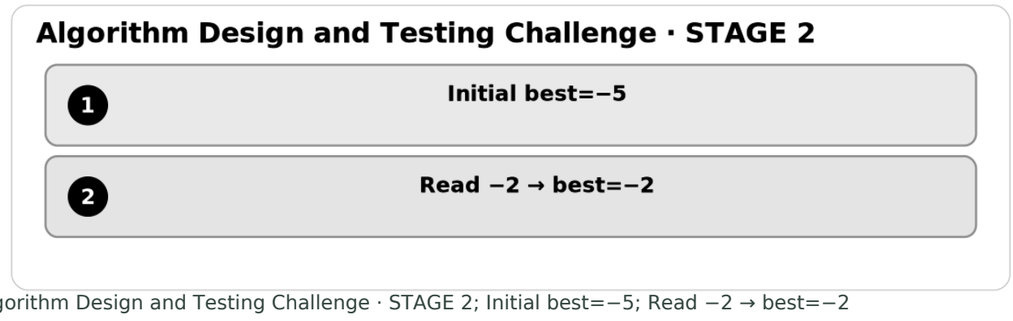
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Algorithm Design and Testing Challenge · STAGE 1; Initial best=−5

New Learning



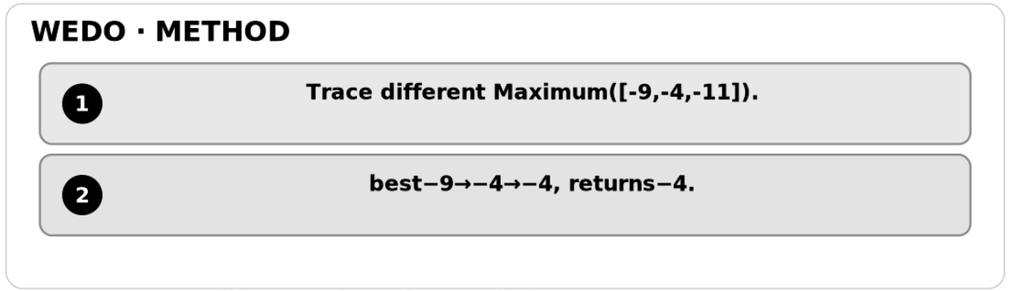
Algorithm Design and Testing Challenge · STAGE 2; Initial best=−5; Read −2 → best=−2

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace different Maximum([-9,-4,-11]).

Starting best at0 would be incorrect for all-negative arrays.



WEDO · METHOD; Trace different Maximum([-9,-4,-11]).; best−9→−4→−4, returns−4.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

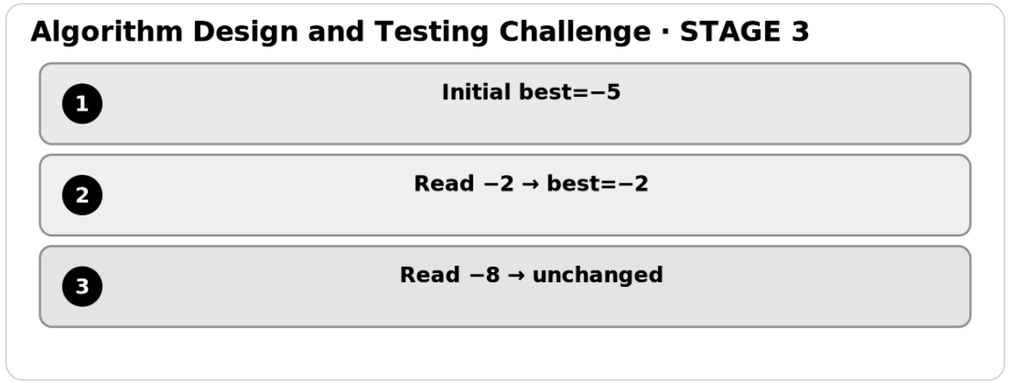
Guided 1 — CHECK

Model response
Starting best at0 would be incorrect for all-negative arrays.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Algorithm Design and Testing Challenge · STAGE 3; Initial best=−5; Read −2 → best=−2; Read −8 → unchanged

New Learning



Algorithm Design and Testing Challenge · STAGE 4; Initial best=−5; Read −2 → best=−2; Read −8 → unchanged; Return −2 (not zero)

I DO — WORKED EXAMPLE

Follow the worked method

Trace Maximum([7,3,10,4]).

Model response

The invariant tracks largest seen, not last examined.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Initialising maximum to zero works for every integer array.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

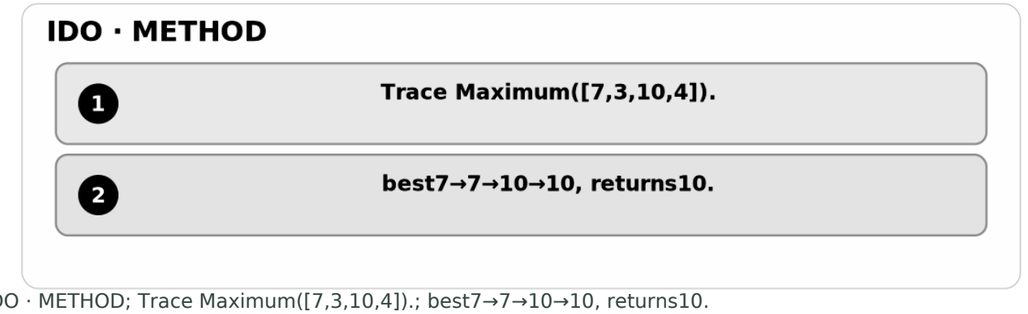
Compare O(n) maximum scan with sorting first and describe useful test coverage. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace Maximum([7,3,10,4]).

The invariant tracks largest seen, not last examined.



IDO · METHOD; Trace Maximum([7,3,10,4]).; best7→7→10→10, returns10.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Trace different Maximum([-9,-4,-11]).

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace Maximum([-5,-2,-8]) showing all changes and final value. [6]

Question 2

Write algorithm for maximum of array, including empty input case. [7]